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CASE REPORT

Sudden death from an epileptic seizure due to capillary telangiectasias in the hippocampus

Yuluo Liu[1] · Yue Liang[1] · Fang Tong[1] · Weisheng Huang[1] · Lopsong Tinzing[1] · Jehane Michael Le Grange[2] · Feixiang Wang[3] · Yiwu Zhou[1]

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Abstract
Cerebral capillary telangiectasia (CCT) is a type of vascular malformation that is incidentally encountered in clinical practice. Diseased vessels are small and usually clinically benign over the course of a patient's life. Although most CCT patients are asymptomatic, the situation becomes complicated when trauma is encountered. A case of sudden death due to an epileptic episode after very mild head trauma is reported, including a retrospective study of 12 cases, to remind peers to pay close attention to CCT especially when located in important functional regions of the brain. After immunohistochemical staining and pathological examination, we speculated that the epileptogenic mechanism of CCT may be similar to that of hippocampal sclerosis. As the definite epileptogenic mechanism of CCT in the hippocampus is still elusive, we suggest that more research should be conducted on CCT.

Keywords Forensic pathology · Cerebral capillary telangiectasia · Natural death · Temporal lobe seizure · Hippocampus

Introduction

Cerebral capillary telangiectasia (CCT) is a type of central nervous system (CNS) vascular malformation consisting of localized collections of multiple, thin-walled vascular channels interspersed within normal neural parenchyma. They are incidentally encountered and are usually clinically benign over the course of a patient's life [1]. In contrast to other vascular malformations of the brain, most CCT cases are discovered incidentally during autopsy as it is rare to encounter symptomatic capillary telangiectasias and difficult to visualize using cerebral angiography and MRI studies. CCT cases are only thought to be symptomatic when the lesions are large, usually with malformations measuring >1.0 cm in dimension. The clinical presentation of CCT cases typically includes seizures, cranial nerve dysfunction, confusion, dizziness, visual changes, vertigo, tinnitus, or progressive spastic paraparesis [2]. Histologically, they are comprised of a collection of dilated capillaries without smooth muscle or elastin, interspersed in normal neural parenchyma, and exert no mass effect on

✉ Feixiang Wang
wangfx@ssfjd.cn

✉ Yiwu Zhou
zhouyiwu@outlook.com

Yuluo Liu
liuyuluo93@sina.com

Yue Liang
yueyueashin@gmail.com

Fang Tong
fytongfang@gmail.com

Weisheng Huang
weisheng@hust.edu.cn

Lopsong Tinzing
m15972938557_2@163.com

Jehane Michael Le Grange
jehanelegrange@gmail.com

1 Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, No. 13 Hangkong Road, Wuhan 430030, People's Republic of China

2 The First Clinical college, Wuhan Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1277 Jiefang Avenue, Wuhan 430022, People's Republic of China

3 National Ministry of Justice, Shanghai Key Laboratory of Forensic Medicine, Institute of Forensic Science, Shanghai 200063, People's Republic of China

adjacent brain tissue. To the best of our knowledge, the lesions have rarely been reported in previous research and therefore the incidence and mortality of CCT is not clear. This paper describes a case of sudden death secondary to capillary telangiectasias in the hippocampus and includes a retrospective study of 12 cases of CCT. The discussion will mainly focus on the mechanism of epilepsy in this case.

Case report

A 54 year old female presented with a headache and dizziness after having sustained a punch to the head. Three hours later, she was transported to the local emergency department by ambulance. Her medical history revealed that she had a lifelong history of mental illness. Physical examination revealed minor abrasions on the head, lips, neck and limbs of the patient. Other medical examinations did not indicate remarkable changes. She experienced dyspnea and discomfort in the neck at about 16 h post-trauma. Without medication, she was soon found dead on the floor of the ward in a prone position. She was cyanotic and there was a significant amount of blood in her mouth. Unfortunately, the exact events of her death were not witnessed.

An autopsy was performed within a few hours of her death. Abrasions and contusions were noted on her head and limbs. In addition to hemorrhaging of the upper gingiva and under the lips, scattered, punctate petechial hemorrhages were also noted within the palpebral conjunctivae.

Dissection of the anterior neck showed no evidence of soft tissue hemorrhage or injuries to the vessels. There were focal areas of atherosclerosis in the aorta and coronary arteries. The lungs were congested and edematous, and the alveolar spaces were filled with significant volumes of pink edematous fluid. Pancreatic hemorrhages were also noted.

CNS examination showed minor herniation of the cerebellar tonsil and the uncus of the hippocampus, which suggested a mild degree of brain swelling. An aggregation of congested vasculature was apparent in the left uncus of the hippocampus and the adjacent amygdala (Fig. 1). The pathologic changes were observed in an area measuring 1.8 cm × 1.2 cm × 1.0 cm.

Microscopic examination revealed a vascular malformation in tissue taken from the left hippocampus (Fig. 2a). Morphologically, there was a collection of dilated capillaries without smooth muscle or elastin, interspersed in the normal neural parenchyma, compatible with cerebral capillary telangiectasia. Some hippocampal pyramidal cells were partially replaced by the dilated capillaries and neuronophagia was obvious (Fig. 2b). Nissl's staining showed that the perilesional pyramidal cells were degenerated or necrotic. Hemorrhagic leakage was also apparent. In addition, glial fibrillary acidic protein (GFAP) staining showed that there was no marked gliosis and mossy fiber sprouting appeared in the pyramidal layer and the dentate gyrus of the hippocampus. Except for the perilesional hemorrhages, there were scattered hemorrhages located in the frontal lobe, midbrain and corpus callosum. Scattered subarachnoid hemorrhages were noted in the frontal lobe.

Aside from the acute hypoxic neuronal changes, mild gliosis, and mild cerebrovascular arteriosclerosis in some regions, no malformations, calcifications, neuronal migration disorders or any destructive, demyelinating, or metabolic lesions were observed in the rest of the brain.

Intoxication was ruled out, as toxicological analysis failed to detect any drugs or unusual substances.

Discussion

CCT is commonly located in the pons, temporal lobe, medulla or caudate nucleus and is seldom symptomatic. We searched PubMed, Web of Science and Medline, including all available years and all languages and used the keywords "CCT or BCT or ICT" and "death or autopsy" and we found 4 necropsy and 8 inpatient reports of cases of CCT that were published from 1982 to 2013 [1, 3–13]. The results are summarized in Table 1. All the cases were written in English. These cases involved 1 male and 11 females aged 4 months to 71 years of age (median of 35 years). In our review, 6 cases presented with focal neurological symptoms, 2 cases presented with seizures and focal neurological symptoms, 3 cases presented with headache and focal neurological symptoms and 1 case presented with no symptoms. Symptomatic CCT may be secondary to hemorrhage into or direct compression onto the adjacent normal brain tissue. The various symptoms may be corelated with the size and location of CCT. Localization of CCT was as follows, in 4 cases CCT occurred in the brainstem, in 3 cases CCT occurred in the cerebellum, in 2 cases CCT occurred in the cerebrum, in 1 case CCT occurred in the hippocampus, in 1 case CCT occurred in both the brainstem and cerebrum and in 1 case CCT occurred in the brainstem, cerebrum and cerebellum. We found that almost all of the CCT located in the cerebellum or the brainstem presented with neurological symptoms and ataxia, CCT in the cerebrum presented with headache, and CCT in the hippocampus presented with mental illness. Additionally, 4 of the cases coexisted with other vascular malformations like cavernous angiomas, arteriovenous malformations and developmental venous anomalies. Previous studies have indicated that CCT and cavernous malformations represent two pathologic extremes within the same vascular malformation category [14].

In our case, the CCT was located in the uncus of the hippocampus and the adjoining amygdala. This location has not commonly been reported in previous research [15]. As the association between epilepsy and the hippocampus has been well studied, vascular malformations of the hippocampus could readily present as seizures and cranial nerve dysfunction

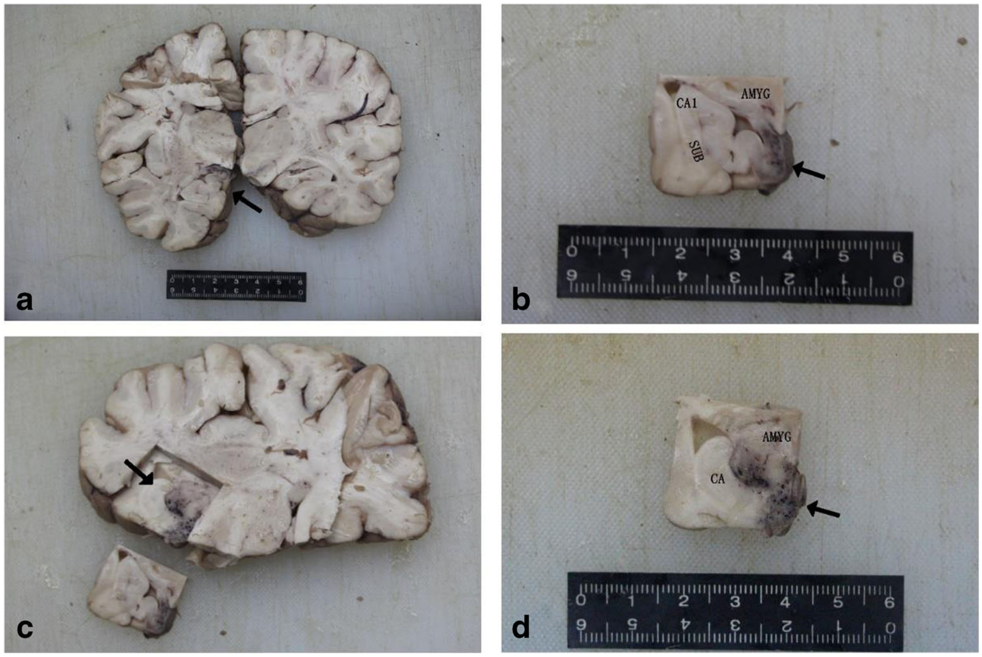

Fig. 1 Vascular malformation involves the left uncus of the hippocampus and the adjacent amygdala. **a** section of the brain showing the left head of the hippocampus (arrows) and (**b**) higher magnification a group of congested vessels distributed in the uncus of the hippocampus (arrows). **c** Section of the head of the hippocampus from the front of A (arrows) which showed (**d**) a group of congested vessels located at the uncus of hippocampus and the adjacent amygdala (arrows)

[16]. Some research even suggests that hippocampal onset accounts for at least 80% of all temporal lobe seizures [17]. A similar case was reported, where a patient with unknown CCT of the hippocampus and a lifelong history of mental illness and mild intellectual disability, died from cardiorespiratory arrest and acute global anoxic/ischemic encephalopathy [3]. The complexity in our case was that the patient had experienced head trauma prior to death, which could result in it being misdiagnosed as a post-traumatic seizure (PTS). However, PTS always presents after brain trauma, which is positively correlated to the incidence. In our case, the medical history and the pathologic examination showed minor brain injury without contusion, and therefore, PTS can be excluded [18, 19]. Since CCT of the hippocampus exists as a potential epileptogenic factor, even a very mild head injury should be emphasized. We hypothesize that the very mild head injury as a predisposing factor combined with the CCT of the hippocampus induced the lethal epileptic seizure. The cause of death was a lethal epileptic seizure triggered by CCT. The manner of death was natural.

The diagnosis of sudden death due to CCT is a challenging task in forensic practice. Unlike other vascular malformation, CCT is seldom symptomatic or ruptured, which often results in lesions being ignored. Additionally, clinical signs and symptoms of CCT vary and are easily confused with epilepsy, focal neurologic deficits, glial tumors or stroke. Besides, the precise pathogenetic and lethal mechanisms of CCT remains unclear, which may lead to a disputed cause of death. Symptoms may present due to hypoperfusion and hypoxic injury, since dilated capillaries may lead to blood stasis. In our case, as a predisposing factor of death, the very mild brain injury may lead to brain swelling/edema and acute hypoxic neuronal changes in

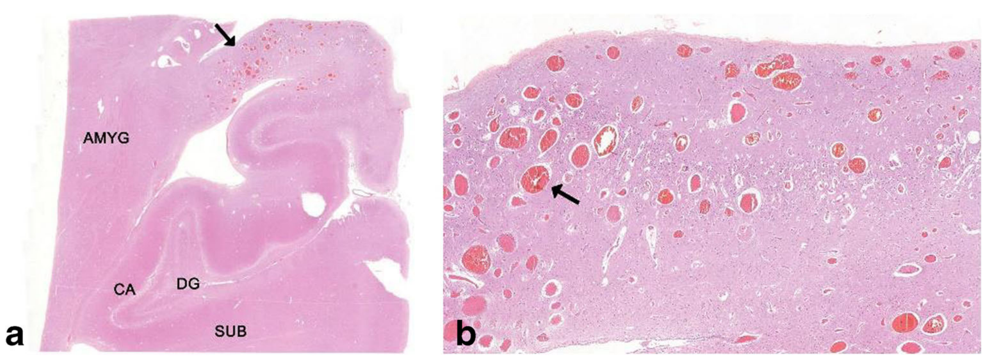

Fig. 2 Pathology features of CCT of the hippocampus. **a** A cluster of vessels aggregated in the uncus of the hippocampus (arrows). H&E 1:6. **b** Higher magnification showed partial neuron loss from the straum oriens to the straum moleculare, caused by the dilated capillaries without smooth muscle or elastin (arrows), distributed to all layers of the cornu ammonis and interspersed in normal neural parenchyma, compatible with a capillary telangiectasia. H&E 1:39

Table 1 Reported cases of CCT (1982–2013)

Authors and Publication Year	Age & Sex	Patient Outcome	Cause of Death	Hemorrhage	CCT Location	Size	Clinical Manifestations	Neuropathologic Findings	Coexisting Lesions
Economou et al. [3]	52/F	Dead	Cardiorespiratory arrest	N	The right posterior subiculum and adjoining portion of the CA1 subfield of the hippocampus	Not mentioned	Mental illness and mild intellectual disability	Dilated vascular channels interspersed in the normal brain,extensive acute global anoxic/ischemic changes,mild subpial cortical gliosis,mild cerebrovascular arteriosclerosis	None
Huddle et al. [4]	5/F	Dead	Natural	N	Midbrain, Pons, medulla	Not mentioned	Febrile seizure,ataxic gait,dysarthria,aphasia,mild conductive hearing loss	Innumerable thin-walled, endothelium-lined vascular channels, marked neuronal loss and chronic astrocytosis	None
Graber et al. [5]	4/F	Dead	hemorrhage in the pons	Y	Pons and left thalamus	Not mentioned	Right hemiparesis, quadriparesis, drowsiness, vomiting, oculomotor palsy	Two clusters of telangiectasias with calcifications, astrocyte gliosis and edema	None
Kuzma et al. [6]	71/M	Dead	Unrelated illness	N	Pons	Not mentioned	/	Numerous small, thin-walled vessels separated by neural parenchymal tissue	Cavernous angioma
Tang et al. [1]	39/F	Survived	/	N	Bilateral cerebrum, cerebellum and brainstem	Not mentioned	Generalized seizure,spastic paraparesis,dysarthria,poor visual acuity	/	None
De Gennaro et al. [7]	31/F	Survived	/	N	The right basal ganglia	2.5 cm	Headache	/	None
Bland et al. [8]	4 m/F	Survived	/	Y	The right cerebellar hemisphere	Not mentioned	High-pitched scream,lethargy,left gaze preference	Cerebellar hematoma, abnormal collection of small caliber vessels within the normal cerebellar parenchyma, mild inflammation	None
Pozzati et al. [9]	10 m/F	Survived	/	N	The right fronto-parietal lobe	7 cm	Left hemiparesis,intermittent vomiting	Several dark-purple nodules,variably sized, thin-walled vascular channels	None
Chang et al. [10]	42/F	Survived	/	N	The right cerebellar hemisphere	5 cm	Occipital headaches, dysarthria, vertigo, nausea, vomiting, diplopia, residual gait ataxia	Capillary telangiectasia, with no elastic tissue noted in the vessel walls	Arteriovenous malformation of the right, posterior inferior cerebellum
Bonneville et al. [11]	52/F	Survived	/	N	Brainstem	Not mentioned	Confusion, distorted visual perceptions, photophobia, neck pain, swallowing problems, and poor balance	/	Cavernous malformations, developmental venous anomalies
Goyal et al. [12]	51/F	Survived	/	N	The central pons	Not mentioned	Sudden onset deafness, weakness and numbness of right arm and leg, slurring of speech and diplopia, truncal ataxia	/	None
McCormick et al. [13]	24/F	Survived	/	Y	The left cerebellar hemisphere	Not mentioned	Left occipital headaches, vertigo, and left-sided incoordination	Acute hemorrhage, numerous vascular channels with astrocyte gliosis, focal collections of	Cavernous malformations, venous angioma

Table 1 (continued)

Authors and Publication Year	Age & Sex	Patient Outcome	Cause of Death	Hemorrhage	CCT Location	Size	Clinical Manifestations	Neuropathologic Findings	Coexisting Lesions
								hemosiderin-bearing macrophages, cavernous malformation and venous angioma was found adjacent to the hematoma	
Current case	54/F	Dead	lethal epileptic seizure	N	The left uncus of hippocampus and adjacent amygdala	1.8 cm × 1.2 cm × 1.0 cm	Mental illness, Headache, dizziness, dyspnea and discomfort of the neck	Mild brain edema, dilated capillaries without smooth muscle or elastin, neuronophagia, necrotic pyramidal cells	None

some extent, which in turn induced the lethal epileptic seizure. Appropriate preventive measures should be taken based on the general characteristics of the clinical data and pathological findings. We should put more emphasis on the common sites of CCT and neuropathologic examination.

To determine the relationship between the lethal seizure and the CCT of the hippocampus, we compared the lesion with various aspects of hippocampal sclerosis (HS). HS is a common pathology encountered in mesial temporal lobe epilepsy, as well as other epilepsy syndromes, in both surgical and post-mortem practice [20]. The relevant epileptogenic characteristics of HS mainly comprises of subfield neuronal loss, gliosis and mossy fiber sprouting. Similarly, the CCT located in the hippocampus caused significant neuronal loss. However, the mossy fiber sprouting and gliosis, a reparative response to hippocampal neuronal loss, was absent in CCT. The difference may be attributed to the fact that CCT exerted no mass effect on the adjacent brain tissue and was not located in the cornu ammonis but rather the uncus of the hippocampus and the adjoining amygdala. We speculated that the epileptogenic mechanism of CCT in our case may be ascribed to neuron loss of the hippocampus. And the CCT in our case is larger than usual, which lead to more neuronal loss and more readily manifested as cognitive deficits and mental illness [21].

Conclusion

In the practice of forensic medicine, sudden death due to CCT always leads to civil disputes. As these cases are rare, it is often difficult to determine the primary cause of death. Since CCT is mostly asymptomatic and elusive in radiologic examination, we must emphasize that a systematic neuropathologic examination is pivotal in the diagnosis of CCT. The autopsy and pathology examination should pay close attention to the common sites of CCT, such as brainstem, cerebellum and hippocampus. Meticulous gross examination and extensive histopathology examination are needed especially when suspicious lesions discovered. After comparing CCT with HS, the potential epileptogenic mechanism of CCT may be due to the significant neuronal loss in the hippocampus, similar to that of HS. Additionally, the exact epileptogenic mechanism of CCT in the hippocampus is still unclear and we suggest that more research should be conducted on CCT.

Key points

1. We report a case of sudden death due to CCT in the hippocampus.
2. The cause of death was a lethal epileptic seizure caused by CCT, which was diagnosed via an autopsy and pathological examination.

3. We reviewed the related characteristic of 12 CCT cases published from 1982 to 2013.
4. The epileptogenic mechanism of CCT in the hippocampus was discussed.

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Compliance with ethical standards

Conflict of interest None.

Ethical approval This study was approved by the Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei, China.

Informed consent Written informed consent was obtained from the family members of the patient.

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